

H1N1 Vaccine Prediction

Business Problems Solved • Logistic Regression Project

This project translates a public-health survey dataset (26,707 respondents, 33 features) into a working prediction system for H1N1 vaccine uptake. Below are the concrete business problems the project addresses, how each was solved, and the practical impact of the solution.

Dataset	Model	Metric	Validation
26,707 rows · 33 features	Logistic Regression	F1-score (imbalanced target)	5-fold Stratified CV

Business Problems & Solutions

1

Who is unlikely to get vaccinated?

PROBLEM

Public health teams had no reliable way to identify, ahead of time, which individuals in the population were least likely to take the H1N1 vaccine, making outreach and awareness campaigns untargeted and inefficient.

SOLUTION

Built a Logistic Regression classifier trained on 26,700+ survey responses covering health behaviors, risk perception, and demographics to predict the probability that a given respondent would get vaccinated.

IMPACT →

Enables targeted outreach: campaigns and reminders can be prioritized toward the segments the model flags as low-likelihood vaccinators.

2

Which factors actually drive vaccination decisions?

PROBLEM

Decision-makers needed to know which behavioral, attitudinal, and demographic factors most influence a person's choice to vaccinate, not just whether they did.

SOLUTION

Used L1-regularized Logistic Regression to rank features by importance (coefficient magnitude), isolating the strongest predictors such as doctor recommendation and perceived vaccine risk/effectiveness.

IMPACT →

Gives program planners evidence-based levers, e.g. reinforcing doctor recommendations, to move vaccination rates instead of guessing at drivers.

3

Incomplete and inconsistent survey data

PROBLEM

Real-world survey data arrived with missing responses and inconsistent value ranges across dozens of columns, which would silently bias or break a downstream model if left untreated.

SOLUTION

Diagnosed missing data with a custom missing-value profiler and applied a column-specific imputation strategy (constant + missing-indicator, or mode + missing-indicator) instead of a one-size-fits-all fill.

IMPACT →

Preserves the signal that a value was missing (which can itself be predictive) while keeping the dataset fully usable for modeling.

4

Outliers skewing statistical assumptions

PROBLEM

Numeric survey fields contained extreme values that could distort a distance- and gradient-sensitive model like Logistic Regression.

SOLUTION

Built an automated outlier detector that chooses Z-score or IQR per column based on its skewness, then quantifies how many outliers exist and at what thresholds.

IMPACT →

Gives a defensible, repeatable outlier-handling process instead of ad hoc manual review of 30+ columns.

5

A rare-event target that a naive model would ignore

PROBLEM

Far fewer people vaccinated than didn't, so a model optimizing for plain accuracy could simply predict "no vaccine" for everyone and still score well while being useless in practice.

SOLUTION

Systematically compared default vs. `class_weight="balanced"` Logistic Regression across multiple scaling strategies, optimizing for F1-score on the vaccinated (positive) class rather than raw accuracy.

IMPACT →

Produces a model that actually surfaces likely vaccinators/non-vaccinators, not one that hides behind class imbalance.

6

A default 0.5 cutoff wasn't the right decision rule

PROBLEM

Logistic Regression's standard 0.5 probability threshold is not automatically the best cutoff for an imbalanced target, leaving performance on the table.

SOLUTION

Used out-of-fold cross-validated probabilities to search for the F1-optimal decision threshold, then validated that threshold on a held-out test set.

IMPACT →

Squeezes additional, measurable F1 performance out of the same model with no extra data, purely through a smarter decision rule.

7

Choosing a trustworthy model configuration

PROBLEM

With multiple scaling methods, class-weight settings, feature subsets, and engineered interaction terms all plausible, there was no structured way to compare configurations or catch overfitting before deployment.

SOLUTION

Built a reusable cross-validation evaluation utility that logs mean/standard deviation of accuracy, precision, recall, and F1 for every configuration into one comparable results table, and checked the train/validation gap for overfitting.

IMPACT →

Turns model selection into an auditable, data-driven comparison rather than a one-off, unverified choice.